# FP8 MQA-Logits Kernel

## DeepGEMM's scoring kernel for DSA's lightning indexer

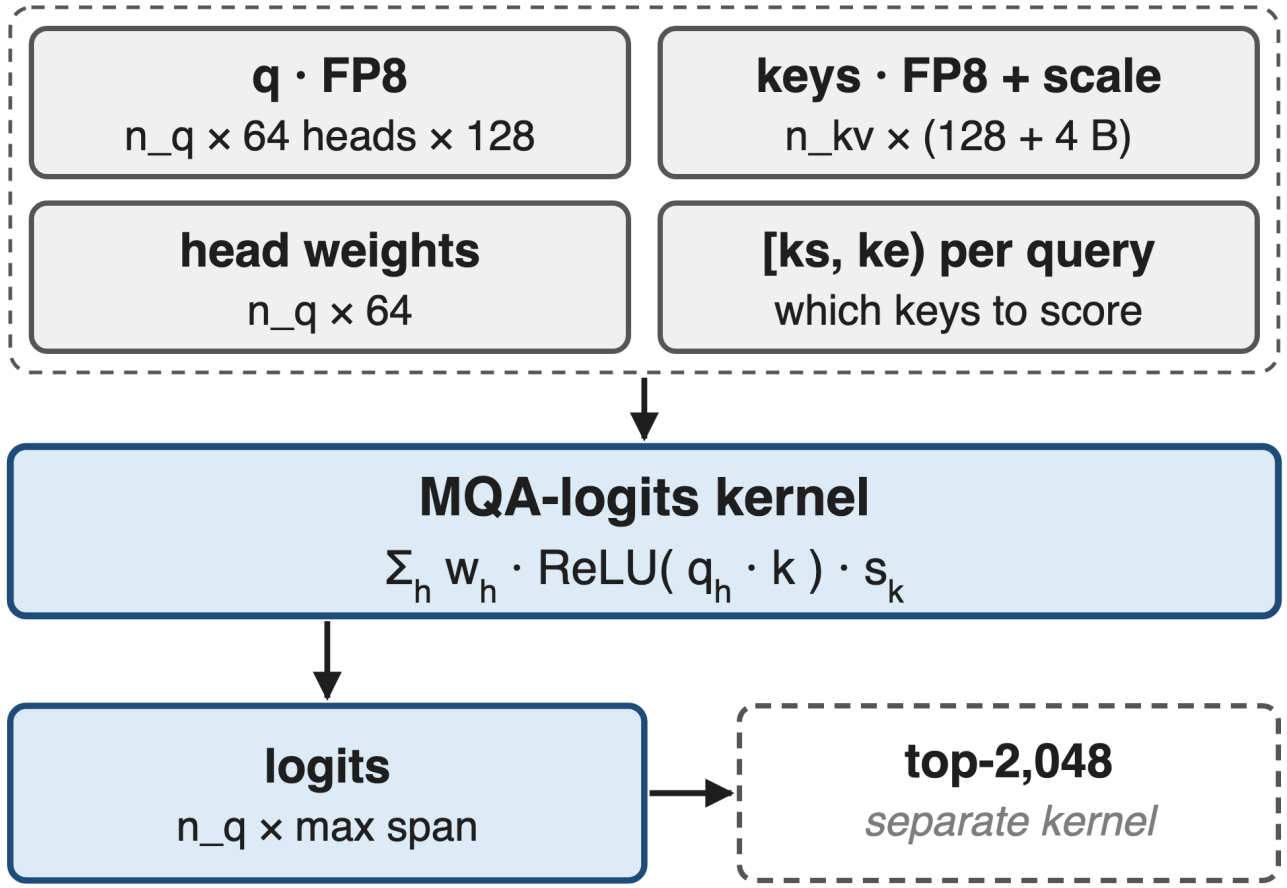
DeepSeek-AI · DeepGEMM (indexer kernels since DeepSeek-V3.2)

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Attention Kernels · github.com/deepseek-ai/DeepGEMM

# The indexer is one MQA-shaped kernel

- Per query: **64** heads × **128** dims, FP8
- One FP8 key per token, shared by all heads
- Shipped with V3.2: **Sep 2025**, for H800



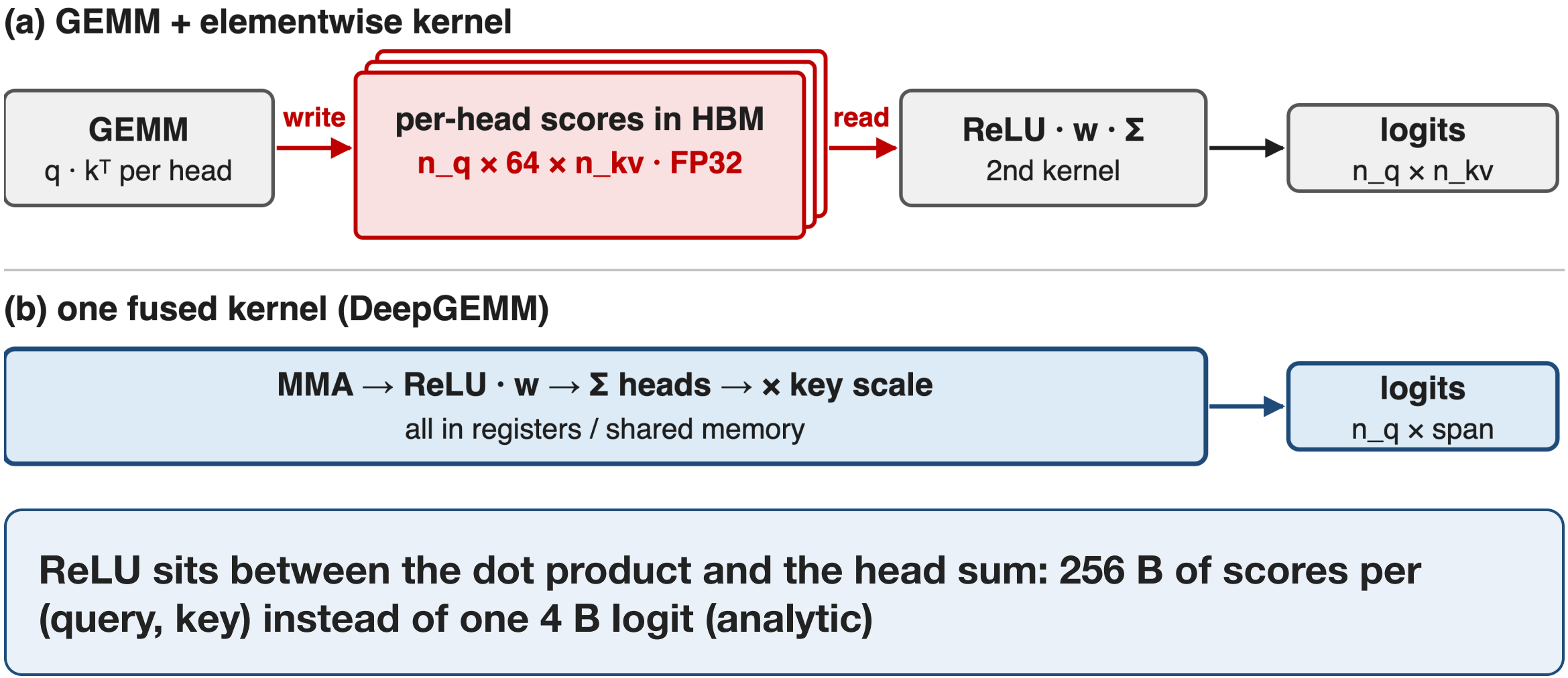
# At 200K, scoring is 81% of prefill time

**81%**

of prefill time goes to the lightning indexer at 200K tokens (27% at 10K): the hot loop of long-context DSA

*IndexCache, arXiv 2603.12201, §1 figure; 30B DSA model on H100 with SGLang*

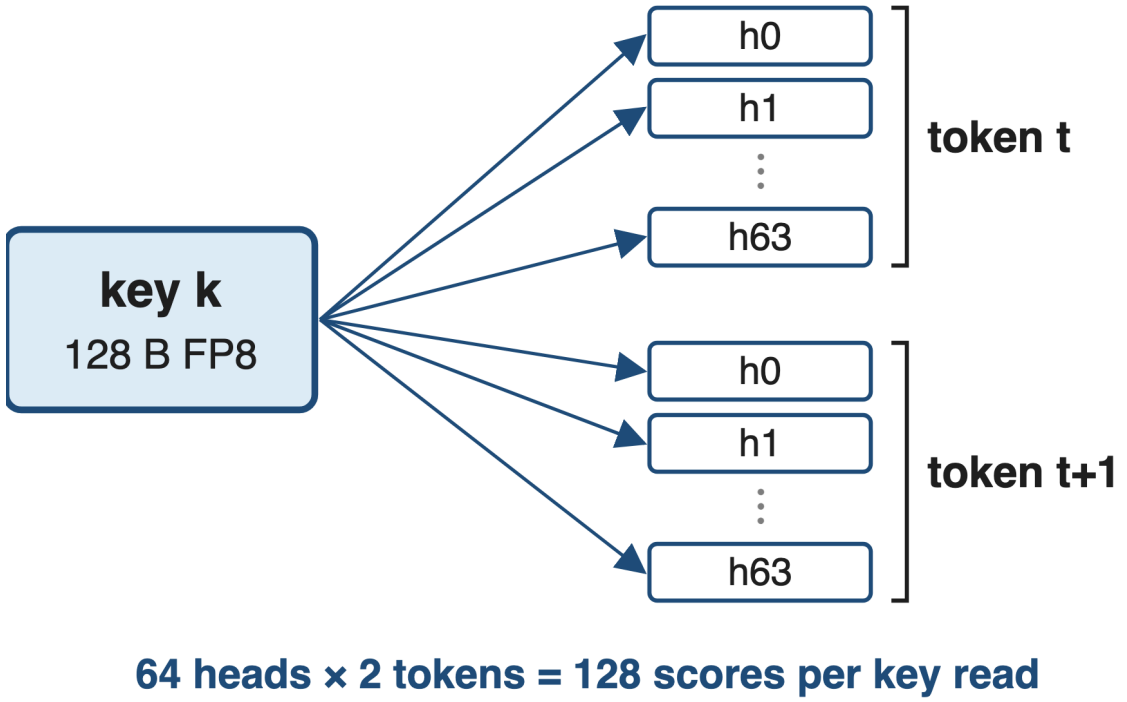
# GEMM + epilogue kernels move 64× the bytes




**ReLU sits between the dot product and the head sum: 256 B of scores per (query, key) instead of one 4 B logit (analytic)**

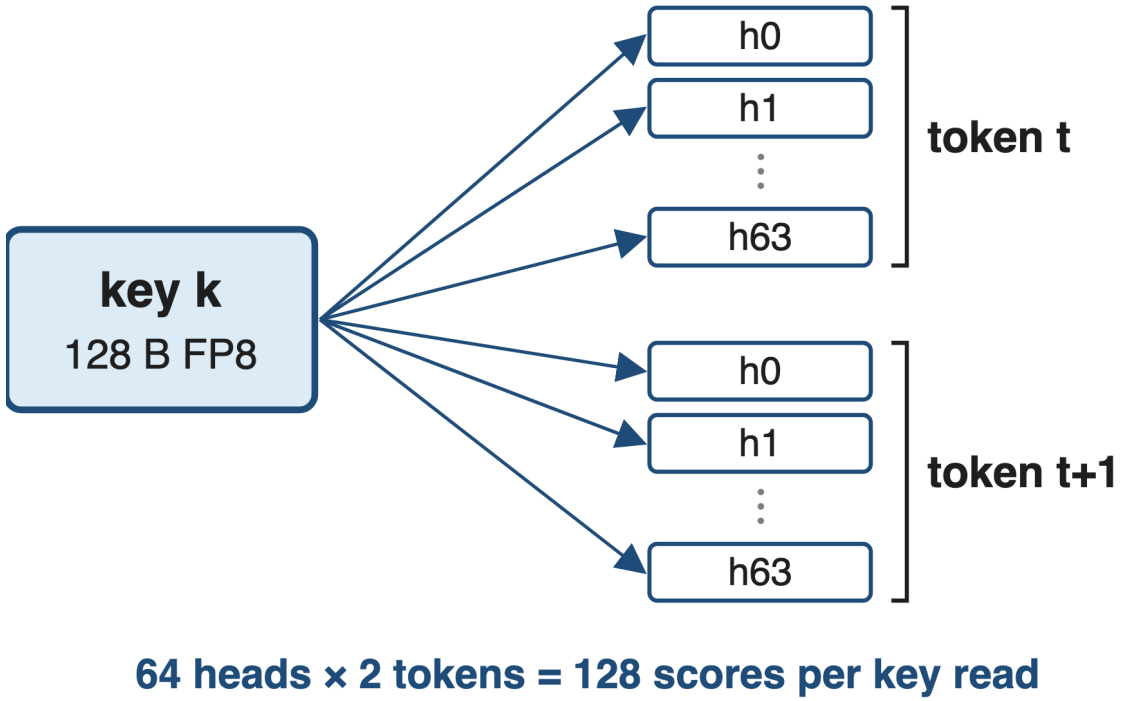
# Insight: one key serves every head and token

- MQA: **1** key read feeds **64** heads × several queries
- K = **128** only: short dot products



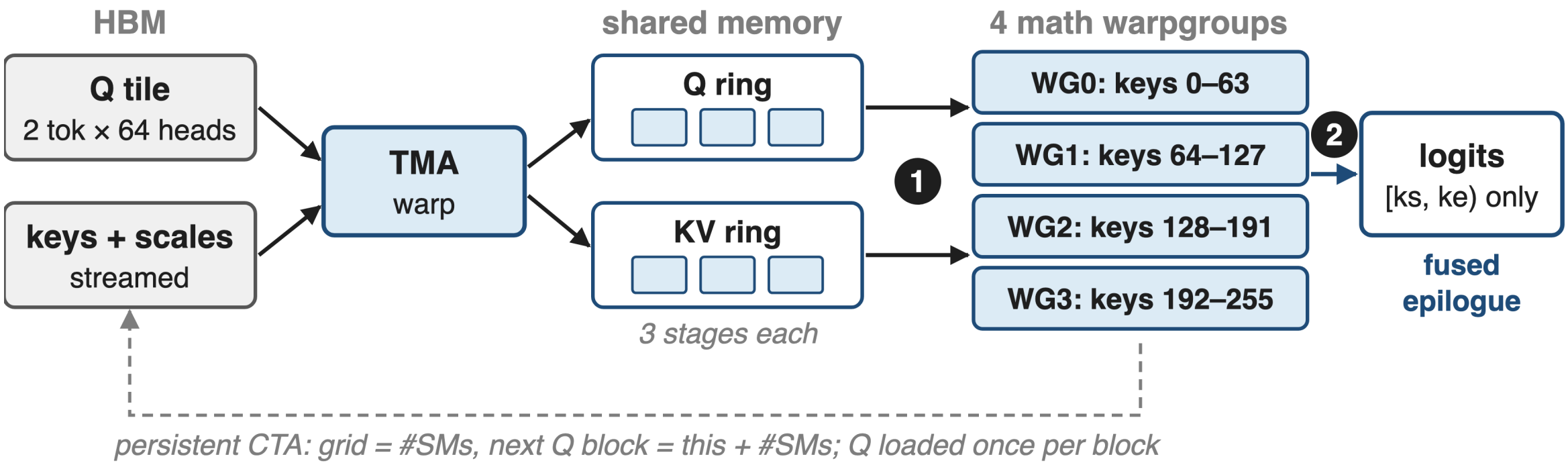
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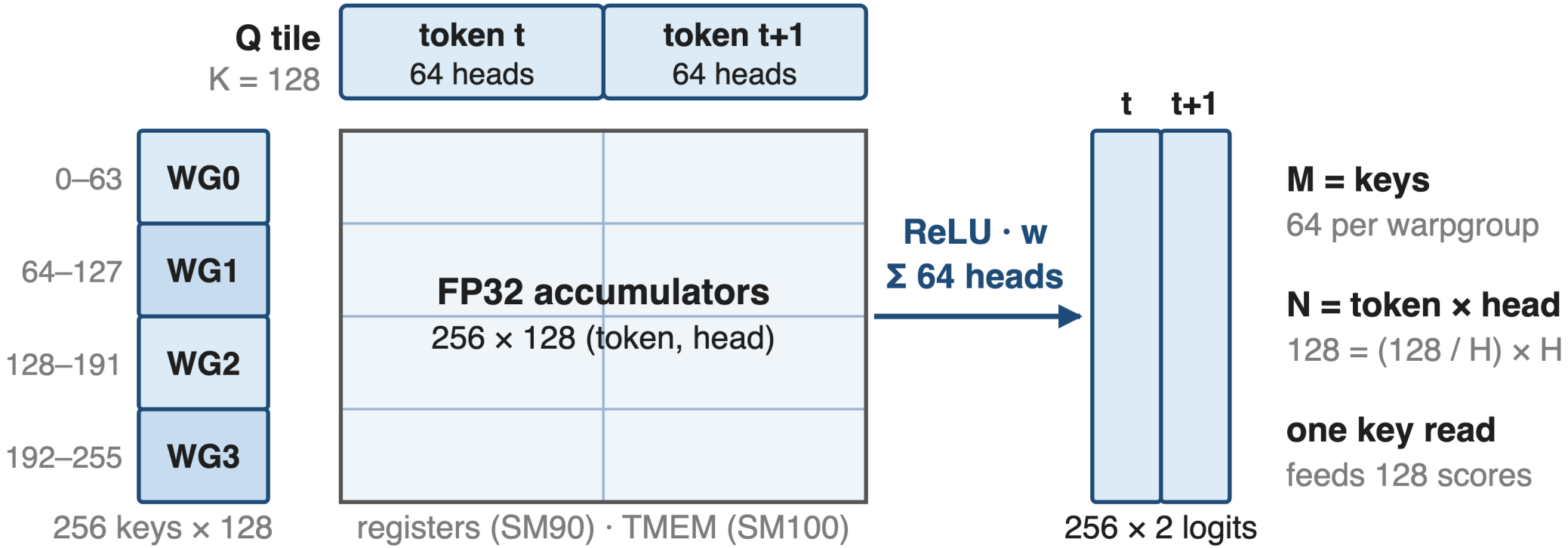

**Put (token × head) on the MMA's N side and finish ReLU · w · Σ in registers: only logits reach HBM**

# One persistent CTA per SM streams keys




*SM90 non-paged kernel: 1 TMA warp + 4 math warpgroups (128 + 512 threads). ❶ tile mapping · ❷ fused epilogue · ❸ paged decode reuses the core*

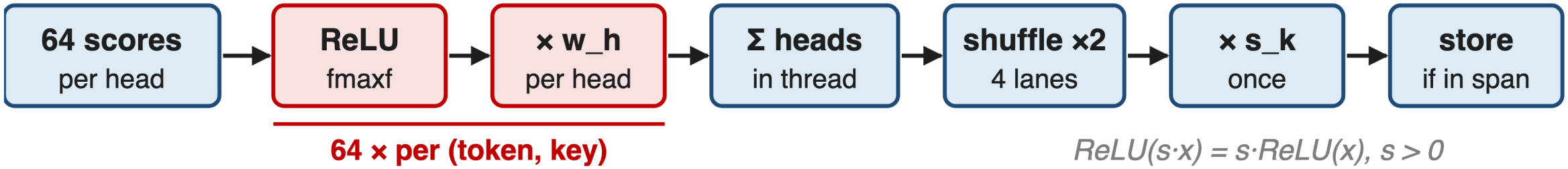
# ❶ 128 (token, head) rows meet 256 keys




**Q tile = 128 / H tokens × H heads: 2 tokens at H = 64, 4 at H = 32; Blackwell uses 384-key splits in TMEM**

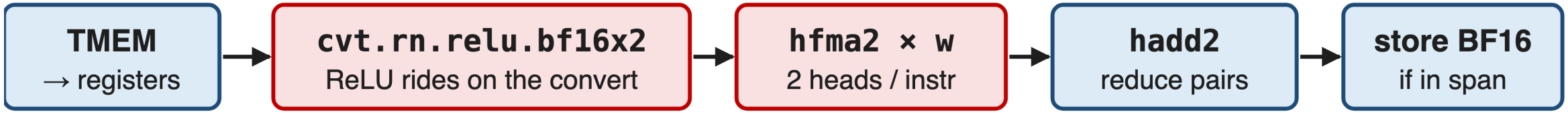
# ❷ ReLU, weights and head sum stay in registers

**(a) Hopper (SM90): FP32 registers**




$ReLU(s \cdot x) = s \cdot ReLU(x),\ s > 0$

**(b) Blackwell (SM100): TMEM, packed BF16**

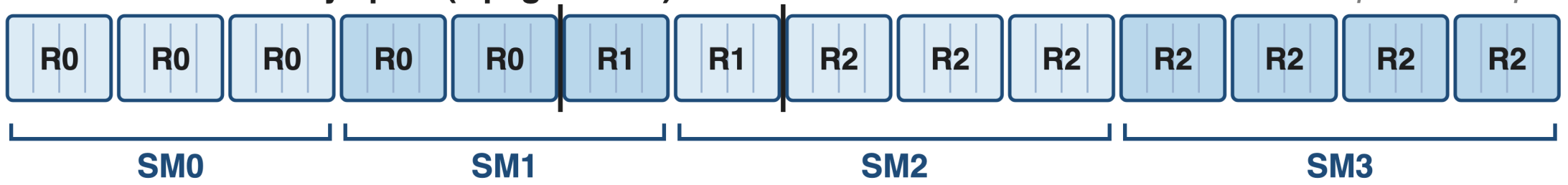



**The per-head steps run 64× per (token, key); DeepGEMM's own test reports this kernel in ReLUs per cycle per SM**

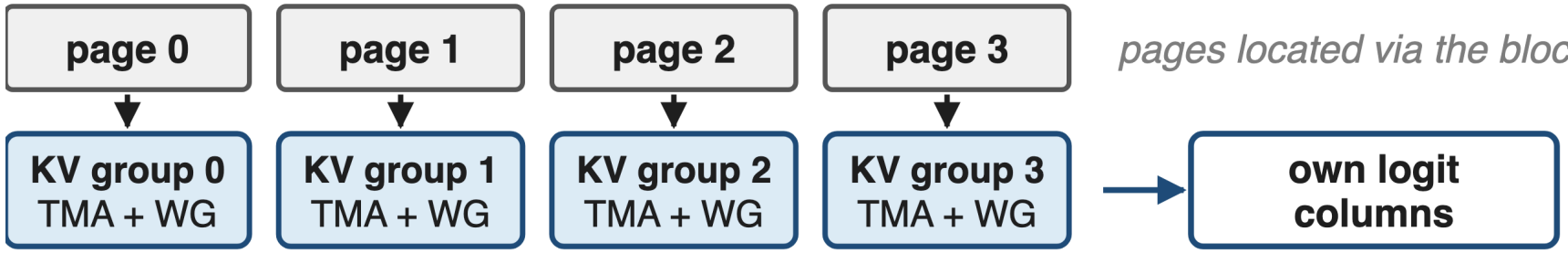
# ❸ Decode splits keys evenly, with no combine

**requests cut into 256-key splits (4 pages of 64)**

*equal shares per SM*




**one split inside a CTA**



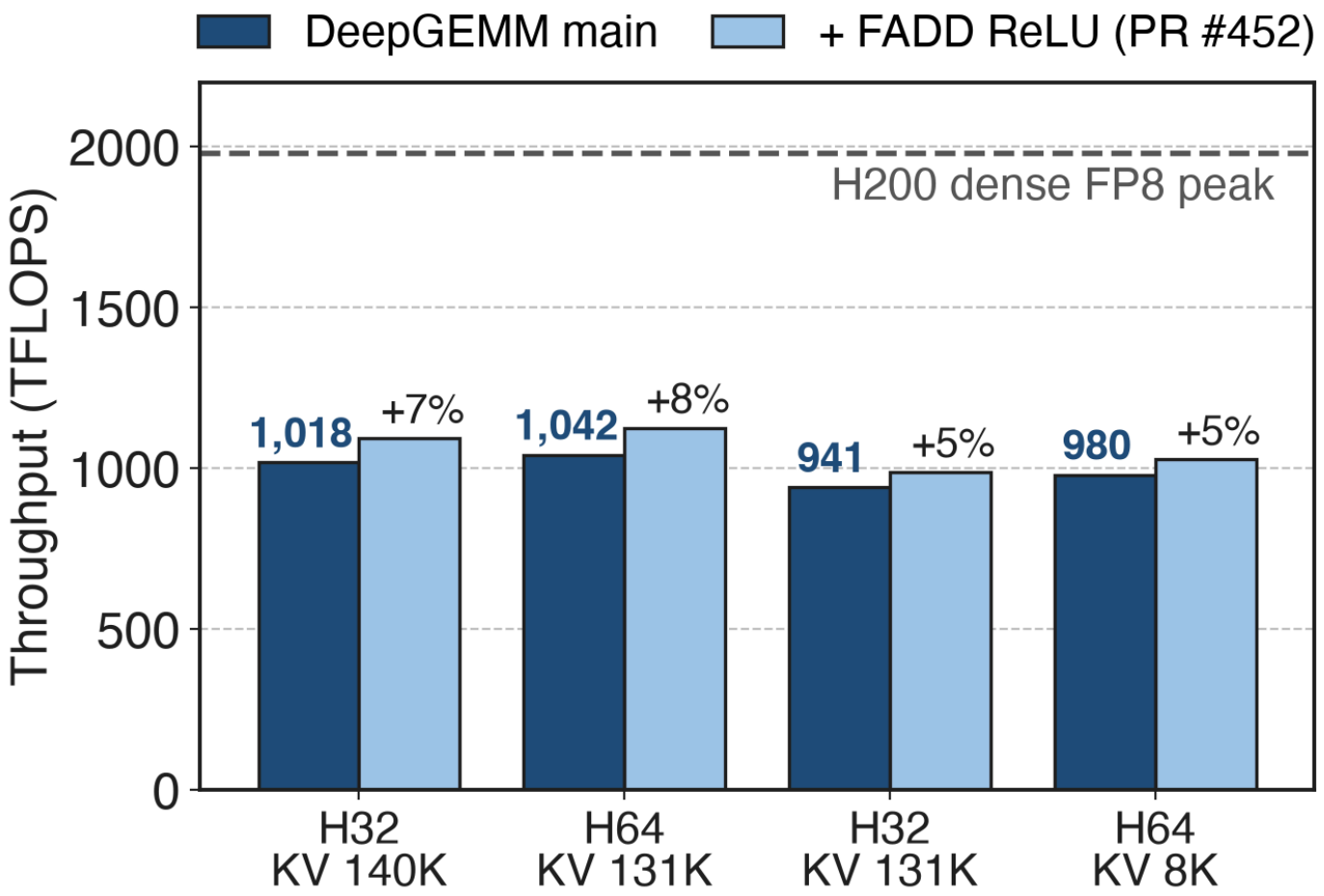

*pages located via the block table*

**no combine step:**
each key is scored independently

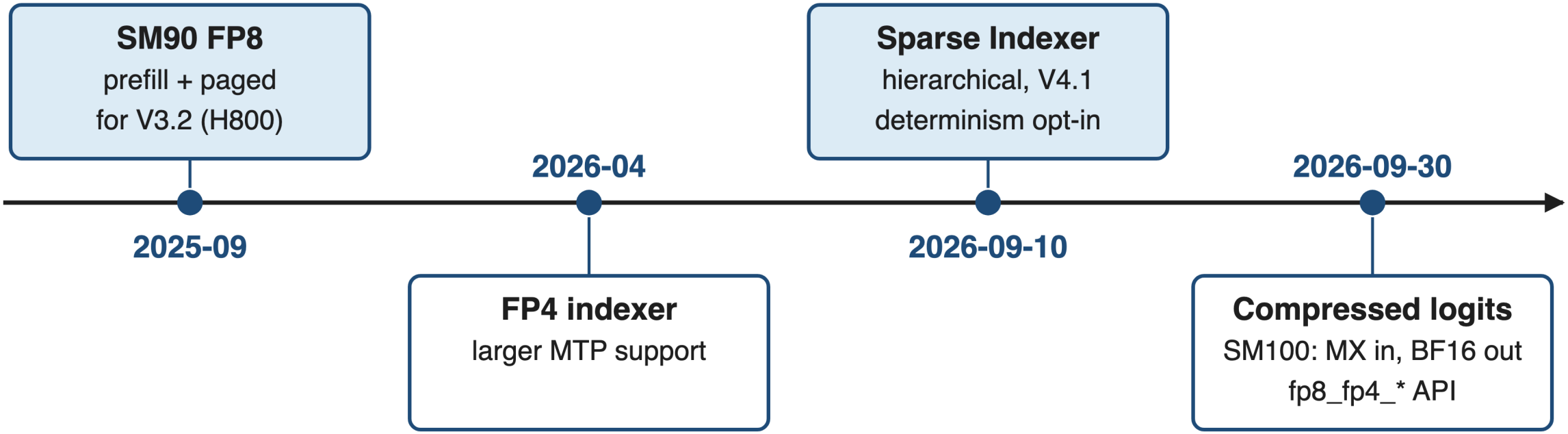
**Decode does 124 FLOP per key byte, under H200's 412 ridge: HBM-bound, so up to 2 MTP tokens share each read (analytic)**

# ~1 PFLOPS FP8 on H200 at production shapes

- **941–1,042** TFLOPS, about **half** of dense peak
- ReLU instruction swap alone: **+5–8%**
- Community data (PR #452), not DeepSeek's



# The kernel family grew FP4 and sparse variants




**MXFP4 keys need 68 B instead of 132 B (1.9× fewer decode bytes, analytic); V4's CSA indexer also runs QK in FP4 (deck 01)**

# The cost: dense logits, masks, determinism

- Logits materialized before a separate top-k
- 4K × 128K chunk: **2 GiB** FP32 (analytic)
- Callers mask past each row's span
- Main branch: SM90 / SM100; SM90: 32 or 64 heads
- Since 2026-09: **non-deterministic** by default

**For RL, set `use_deterministic_algorithms(True)`: GLM-5 saw unstable top-k derail RL (deck 02)**

# Summary

- DSA's indexer: **81%** of prefill time at 200K
- **Insight**: one key serves all heads and tokens: fuse MMA → ReLU · w · Σ
- ❶ 128-row Q tile × 256-key tiles ❷ register epilogue ❸ even split-KV decode
- **~1 PFLOPS** FP8 on H200 (community data); no score tensor in HBM

**https://github.com/deepseek-ai/DeepGEMM**